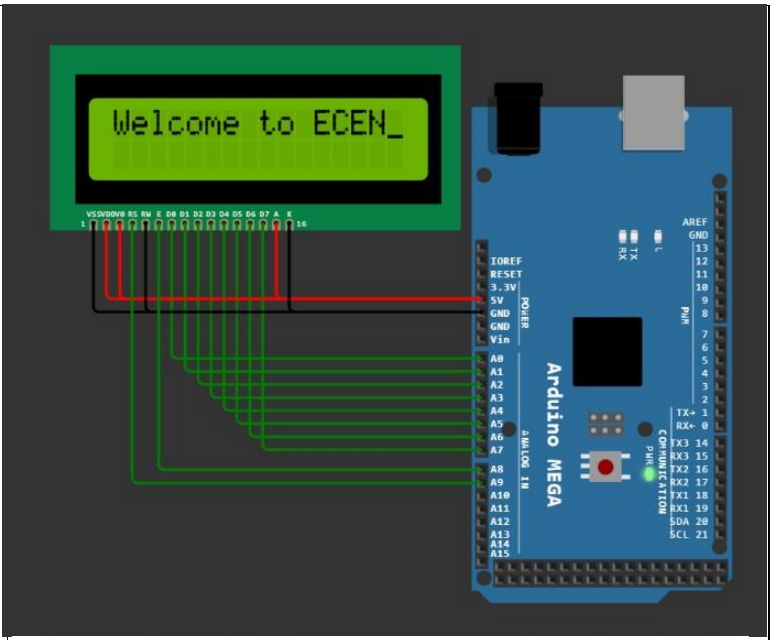
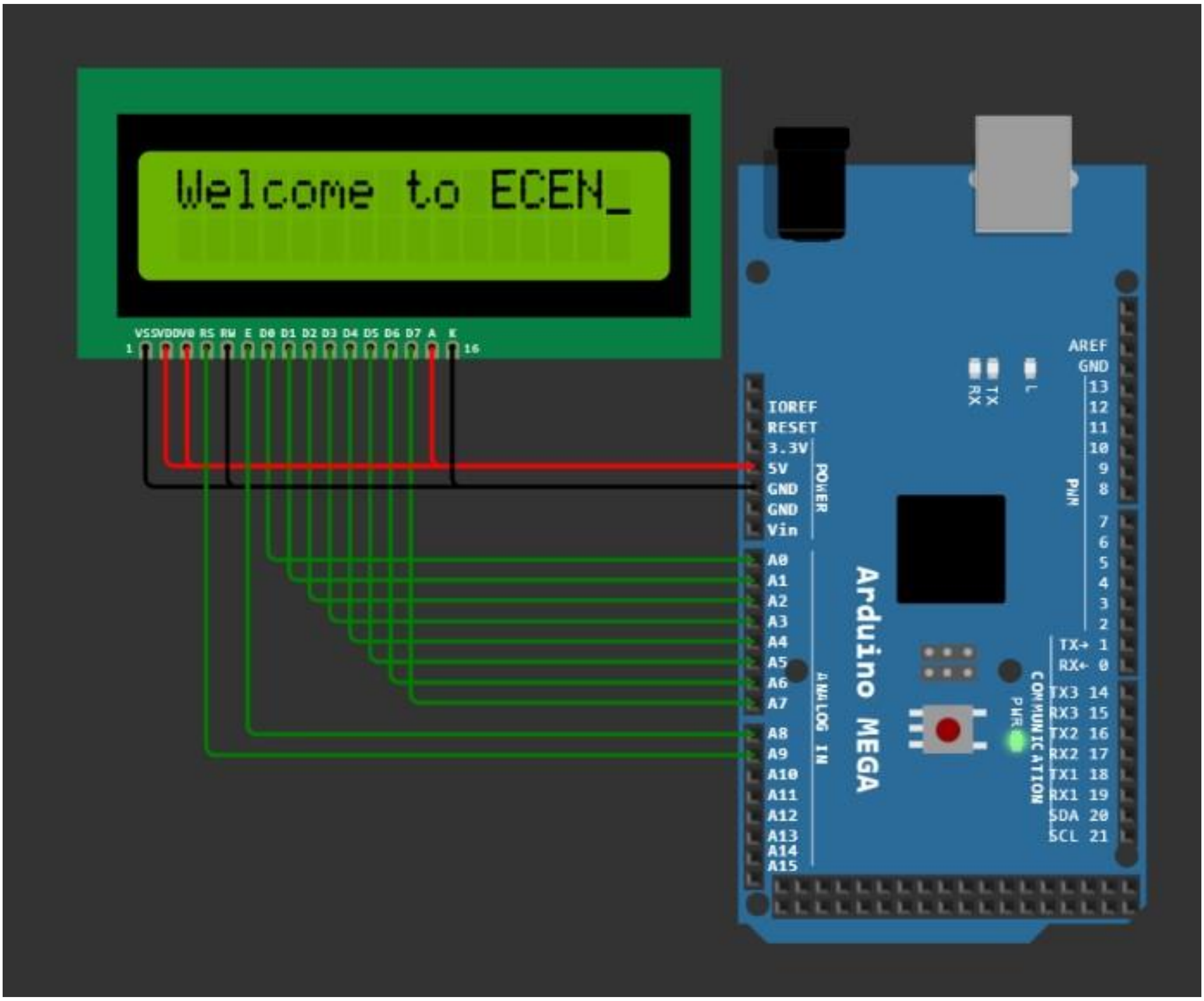
ESP – LCD Display Programming (Lecture) :-

```
volatile char *dirf, *outf, *dirk, *outk;

void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  init_port();
  init_lcd();
  lcd_data("Welcome to ECEN");
}

void lcd_data(volatile char *ptr){
while(*ptr){
    *outf=*ptr;
    *outk=2;  delay1(1);  // rs=1; en=0;
    *outk=3;  delay1(1);  // rs=1; en=1;
    *outk=2;  delay1(1);  // rs=1; en=0;
    ptr++;
  }
}
void init_lcd(){
  lcd_cmd(0x38);  // 8 bit bus & 2 rows function set
  lcd_cmd(0x0f);  // display on, cursor blink
  lcd_cmd(0x01);  // clear display
  lcd_cmd(0x06);  // cursor auto increment
}
void lcd_cmd(volatile char cmd){
  *outf=cmd;
  *outk=0;  delay1(1);  // rs=0; en=0;
  *outk=1;  delay1(1);  // rs=0; en=1;
  *outk=0;  delay1(1);  // rs=0; en=0;
}
void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}

void delay1(volatile int time){
  volatile long i;
  while(time--) for(i=0; i<100; i++);
}

void loop() {}
```



Sr.No.	Hex Code	Command to LCD instruction Register
1	01	Clear display screen
2	02	Return home
3	04	Decrement cursor (shift cursor to left)
4	06	Increment cursor (shift cursor to right)
5	05	Shift display right
6	07	Shift display left
7	08	Display off, cursor off
8	0A	Display off, cursor on
9	0C	Display on, cursor off
10	0E	Display on, cursor blinking
11	0F	Display on, cursor blinking
12	10	Shift cursor position to left
13	14	Shift the cursor position to the right
14	18	Shift the entire display to the left
15	1C	Shift the entire display to the right
16	80	Force cursor to the beginning (1st line)
17	C0	Force cursor to the beginning (2nd line)
18	38	2 lines and 5×7 matrix



RS:- RS is the register select pin. We need to set it to 1, if we are sending some data to be displayed on LCD. And we will set is to 0 if we are sending some comm instruction like clear the screen (hex code 01).

RW:- 0 - Write operation, 1 - Read operation. We only use to write operation, so, it would be ~~gou~~ grounded.

E:- This pin is used to enable the module when a high to low pulse is given to it. A pulse of 450 ns should be given. That transition from High to Low makes the Module ENABLE.

ESP – LCD Display (16 x 2) Set - 1

01 Display "Welcome" on line1. Left Justified :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);


void setup(){
  init_port();
  init_lcd();
  lcd_data("Welcome");
}

void lcd_data(volatile char *ptr){

while(*ptr){
    *outf=*ptr;
    *outk=2;  delay1(1);  // rs=1; en=0;
    *outk=3;  delay1(1);  // rs=1; en=1;
    *outk=2;  delay1(1);  // rs=1; en=0;
    ptr++;
  }
}

void init_lcd(){
  lcd_cmd(0x38);  // 8 bit bus & 2 rows function set
  lcd_cmd(0x0f);  // display on, cursor blink
  lcd_cmd(0x01);  // clear display
  lcd_cmd(0x06);  // cursor auto increment
}

void lcd_cmd(volatile char cmd){
  *outf=cmd;

  *outk=0;  delay1(1);  // rs=0; en=0;
  *outk=1;  delay1(1);  // rs=0; en=1;
  *outk=0;  delay1(1);  // rs=0; en=0;
}

void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}

void loop() {}
```

02 Display "Welcome" on line1. Right Justified:-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);


void setup(){
  init_port();
  init_lcd();
  lcd_data("Welcome");
}

void lcd_data(volatile char *ptr){

while(*ptr){
    *outf=*ptr;
    *outk=2;  delay1(1);  // rs=1; en=0;
    *outk=3;  delay1(1);  // rs=1; en=1;
    *outk=2;  delay1(1);  // rs=1; en=0;
    ptr++;
  }
}

void init_lcd(){
  lcd_cmd(0x38);  // 8 bit bus & 2 rows function set
  lcd_cmd(0x0f);  // display on, cursor blink
  lcd_cmd(0x01);  // clear display
  lcd_cmd(0x06);  // cursor auto increment
  lcd_cmd(0x89);  // set cursor to row 0 col 9
}

void lcd_cmd(volatile char cmd){
  *outf=cmd;
  *outk=0;  delay1(1);  // rs=0; en=0;
  *outk=1;  delay1(1);  // rs=0; en=1;
  *outk=0;  delay1(1);  // rs=0; en=0;
}
void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}

void loop() {}
```

03 Display “Welcome" on line1. Middle :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
 init_port();
 init_lcd();
 lcd_data("Welcome");
}

void lcd_data(volatile char *ptr){

while(*ptr){
   *outf=*ptr;
   *outk=2;  delay1(1);  // rs=1; en=0;
   *outk=3;  delay1(1);  // rs=1; en=1;
   *outk=2;  delay1(1);  // rs=1; en=0;
   ptr++;
 }
}
void init_lcd(){
 lcd_cmd(0x38);  // 8 bit bus & 2 rows function set
 lcd_cmd(0x0f);  // display on, cursor blink
 lcd_cmd(0x01);  // clear display
 lcd_cmd(0x06);  // cursor auto increment
 lcd_cmd(0x84);  // set cursor to row 0 col 4
}

void lcd_cmd(volatile char cmd){
 *outf=cmd;
 *outk=0;  delay1(1);  // rs=0; en=0;
 *outk=1;  delay1(1);  // rs=0; en=1;
 *outk=0;  delay1(1);  // rs=0; en=0;
}

void init_port(){
 dirf=0x30; *dirf=0xff;
 dirk=0x107; *dirk=0x03;
 outf=0x31; outk=0x108;
}

void delay1(volatile int time){
 volatile long i;
 while(time--)  for(i=0; i<100; i++);
}

void loop() {}
```

04 Display “Welcome" on line2. Left Justified :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
 init_port();
 init_lcd();
 lcd_data("Welcome");
}

void lcd_data(volatile char *ptr){

while(*ptr){
   *outf=*ptr;
   *outk=2;  delay1(1);  // rs=1; en=0;
   *outk=3;  delay1(1);  // rs=1; en=1;
   *outk=2;  delay1(1);  // rs=1; en=0;
   ptr++;
 }
}
void init_lcd(){
 lcd_cmd(0x38);  // 8 bit bus & 2 rows function set
 lcd_cmd(0x0f);  // display on, cursor blink
 lcd_cmd(0x01);  // clear display
 lcd_cmd(0x06);  // cursor auto increment
 lcd_cmd(0xc0);  // set cursor to row 1 col 0
}

void lcd_cmd(volatile char cmd){
 *outf=cmd;
 *outk=0;  delay1(1);  // rs=0; en=0;
 *outk=1;  delay1(1);  // rs=0; en=1;
 *outk=0;  delay1(1);  // rs=0; en=0;
}

void init_port(){
 dirf=0x30; *dirf=0xff;
 dirk=0x107; *dirk=0x03;
 outf=0x31; outk=0x108;
}

void delay1(volatile int time){
 volatile long i;
 while(time--)  for(i=0; i<100; i++);
}

void loop() {}
```

05 Display “Welcome" on line2. Right Justified:-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  init_port();
  init_lcd();
  lcd_data("Welcome");
}

void lcd_data(volatile char *ptr){

while(*ptr){
    *outf=*ptr;
    *outk=2;  delay1(1);  // rs=1; en=0;
    *outk=3;  delay1(1);  // rs=1; en=1;
    *outk=2;  delay1(1);  // rs=1; en=0;
    ptr++;
  }
}
void init_lcd(){
  lcd_cmd(0x38);  // 8 bit bus & 2 rows function set
  lcd_cmd(0x0f);  // display on, cursor blink
  lcd_cmd(0x01);  // clear display
  lcd_cmd(0x06);  // cursor auto increment
  lcd_cmd(0xc9);  // set cursor to row 1 col 9
}

void lcd_cmd(volatile char cmd){
  *outf=cmd;
  *outk=0;  delay1(1);  // rs=0; en=0;
  *outk=1;  delay1(1);  // rs=0; en=1;
  *outk=0;  delay1(1);  // rs=0; en=0;
}

void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}

void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}

void loop() {}
```

06 Display “Welcome" on line2. Middle :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  init_port();
  init_lcd();
  lcd_data("Welcome");
}

void lcd_data(volatile char *ptr){

while(*ptr){
    *outf=*ptr;
    *outk=2;  delay1(1);  // rs=1; en=0;
    *outk=3;  delay1(1);  // rs=1; en=1;
    *outk=2;  delay1(1);  // rs=1; en=0;
    ptr++;
  }
}
void init_lcd(){
  lcd_cmd(0x38);  // 8 bit bus & 2 rows function set
  lcd_cmd(0x0f);  // display on, cursor blink
  lcd_cmd(0x01);  // clear display
  lcd_cmd(0x06);  // cursor auto increment
  lcd_cmd(0xc4);  // set cursor to row 1 col 4
}

void lcd_cmd(volatile char cmd){
  *outf=cmd;
  *outk=0;  delay1(1);  // rs=0; en=0;
  *outk=1;  delay1(1);  // rs=0; en=1;
  *outk=0;  delay1(1);  // rs=0; en=0;
}

void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}

void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}

void loop() {}
```

07 Display "Welcome" on line1. Middle & Display "To ECEN Academy" on line2 Middle :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  init_port();
  init_lcd();
  lcd_cmd(0x84);  // set cursor to row 0 col 4
  lcd_data("Welcome");
  lcd_cmd(0xc0);  // set cursor to row 1 col 0
  lcd_data("to ECEN Academy");
}

void lcd_data(volatile char *ptr){

while(*ptr){
    *outf=*ptr;
    *outk=2;  delay1(1);  // rs=1; en=0;
    *outk=3;  delay1(1);  // rs=1; en=1;
    *outk=2;  delay1(1);  // rs=1; en=0;
    ptr++;
  }
}
void init_lcd(){
  lcd_cmd(0x38);  // 8 bit bus & 2 rows function set
  lcd_cmd(0x0f);  // display on, cursor blink
  lcd_cmd(0x01);  // clear display
}

void lcd_cmd(volatile char cmd){
  *outf=cmd;
  *outk=0;  delay1(1);  // rs=0; en=0;
  *outk=1;  delay1(1);  // rs=0; en=1;
  *outk=0;  delay1(1);  // rs=0; en=0;
}

void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}

void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}

void loop() {}
```

08 Display "Welcome" on line1. Enter from Right. Left Justified :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  volatile short a;
  volatile long i;
  init_port();
  init_lcd();
 for(a=0x8f; a>=0x80; a--){
  lcd_cmd(0x01);
  lcd_cmd(a);
  lcd_data("Welcome");
  for(i=0; i<100000; i++);
 }
}
void lcd_data(volatile char *ptr){
while(*ptr){
    *outf=*ptr;
    *outk=2;  delay1(1);  // rs=1; en=0;
    *outk=3;  delay1(1);  // rs=1; en=1;
    *outk=2;  delay1(1);  // rs=1; en=0;
    ptr++;
  }
}
void init_lcd(){
  lcd_cmd(0x38);  // 8 bit bus & 2 rows function set
  lcd_cmd(0x0f);  // display on, cursor blink
  lcd_cmd(0x01);  // clear display
}
void lcd_cmd(volatile char cmd){
  *outf=cmd;
  *outk=0;  delay1(1);  // rs=0; en=0;
  *outk=1;  delay1(1);  // rs=0; en=1;
  *outk=0;  delay1(1);  // rs=0; en=0;
}
void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}

void loop() {}
```

09 Display "Welcome" on line2. Enter from Left. Right Justified :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  volatile short a;
  volatile long i;
  init_port();
  init_lcd();
 for(a=0xb9; a<=0xc9; a++){
  lcd_cmd(0x01);
  lcd_cmd(a);
  lcd_data("Welcome");
  for(i=0; i<100000; i++);
 }
}
void lcd_data(volatile char *ptr){
while(*ptr){
    *outf=*ptr;
    *outk=2;  delay1(1);  // rs=1; en=0;
    *outk=3;  delay1(1);  // rs=1; en=1;
    *outk=2;  delay1(1);  // rs=1; en=0;
    ptr++;
  }
}
void init_lcd(){
  lcd_cmd(0x38);  // 8 bit bus & 2 rows function set
  lcd_cmd(0x0f);  // display on, cursor blink
  lcd_cmd(0x01);  // clear display
}
void lcd_cmd(volatile char cmd){
  *outf=cmd;
  *outk=0;  delay1(1);  // rs=0; en=0;
  *outk=1;  delay1(1);  // rs=0; en=1;
  *outk=0;  delay1(1);  // rs=0; en=0;
}
void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}

void loop() {}
```

10 Display "Welcome" on Line 1. Enter from Left to Middle. Display "To ECEN Academy" on Line2 Enter from Right to Middle :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  volatile short a;
  volatile long i;
  init_port();  init_lcd();

for(a=0x80; a<0x8c; a++){
    lcd_cmd(0x01);
    lcd_cmd(0x04);  // for left shift the cursor;
    lcd_cmd(a);
    lcd_data("emocleW");
    for(i=0; i<100000; i++);
  }
  for(a=0xd0; a>=0xc1; a--){
    lcd_cmd(0x01);
    lcd_cmd(0x85);
    lcd_data("Welcome");
    lcd_cmd(a);
    lcd_data("To ECEN Academy");
    for(i=0; i<100000; i++);
  }  }
void lcd_data(volatile char *ptr){
while(*ptr){
    *outf=*ptr;    *outk=2;  delay1(1);
    *outk=3;  delay1(1);    *outk=2;  delay1(1);
    ptr++;
  }  }
void init_lcd(){
  lcd_cmd(0x38); lcd_cmd(0x0f); lcd_cmd(0x01);
}
void lcd_cmd(volatile char cmd){
  *outf=cmd;    *outk=0;  delay1(1);
  *outk=1;  delay1(1);    *outk=0;  delay1(1);
}
void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}
void loop() {}
```

ESP – LCD Display (16 x 2) Set - 2

01 Display "Welcome to" on line1. Rotate left to right continuously :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  volatile char a;
  volatile long i;
  init_port();
  init_lcd();
while(1){
  lcd_cmd(0x0a);  // Display off cursor on
  lcd_data("Welcome To");
  lcd_cmd(0x0c);  // Display on cursor off
// shifting entire display left 10 position
  for(a=0; a<10; a++)  lcd_cmd(0x18);

  for(a=0; a<25; a++){
    lcd_cmd(0x1c);
    for(i=0; i<100000; i++);
  }
  lcd_cmd(0x01);
 } }
void lcd_data(volatile char *ptr){
while(*ptr){
    *outf=*ptr;   *outk=2;  delay1(1);  // rs=1; en=0;
    *outk=3;  delay1(1);    *outk=2;  delay1(1);
    ptr++;
  } }
void init_lcd(){
  lcd_cmd(0x38);  // 8 bit bus & 2 rows function set
  lcd_cmd(0x0f);  // display on, cursor blink
  lcd_cmd(0x01);  // clear display
}
void lcd_cmd(volatile char cmd){
  *outf=cmd;  *outk=0;  delay1(1);  // rs=0; en=0;
  *outk=1;  delay1(1);    *outk=0;  delay1(1);
}
void init_port(){
  dirf=0x30; *dirf=0xff;  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}
void loop() {}
```

02 Display "Welcome to" on line1. Rotate right to left continuously :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  volatile char a;
  volatile long i;
  init_port();
  init_lcd();
while(1){
  lcd_cmd(0x0a);  // Display off cursor on
  lcd_cmd(0x91);
  lcd_data("Welcome To");
  lcd_cmd(0x0c);  // Display on cursor off

  for(a=0; a<25; a++){
    lcd_cmd(0x18);
    for(i=0; i<100000; i++);
  }
  lcd_cmd(0x01);
 } }
void lcd_data(volatile char *ptr){
while(*ptr){
    *outf=*ptr;   *outk=2;  delay1(1);  // rs=1; en=0;
    *outk=3;  delay1(1);    *outk=2;  delay1(1);
    ptr++;
  } }
void init_lcd(){
  lcd_cmd(0x38);  // 8 bit bus & 2 rows function set
  lcd_cmd(0x0f);  // display on, cursor blink
  lcd_cmd(0x01);  // clear display
}
void lcd_cmd(volatile char cmd){
  *outf=cmd;  *outk=0;  delay1(1);  // rs=0; en=0;
  *outk=1;  delay1(1);    *outk=0;  delay1(1);
}
void init_port(){
  dirf=0x30; *dirf=0xff;  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}
void loop() {}
```

03 Display "ECEN Academy" on line2. Rotate left to right continuously :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  volatile char a;
  volatile long i;
  init_port();
  init_lcd();

while(1){
  lcd_cmd(0x0a);  // Display off cursor on
  lcd_cmd(0xc0);
  lcd_data("ECEN Academy");
// shifting entire display left 10 position
  for(a=0; a<12; a++)  lcd_cmd(0x18);
  lcd_cmd(0x0c);  // Display on cursor off

  for(a=0; a<29; a++){
    lcd_cmd(0x1c);
    for(i=0; i<100000; i++);
  }
  lcd_cmd(1);
 } }
void lcd_data(volatile char *ptr){
while(*ptr){
    *outf=*ptr;   *outk=2;  delay1(1);  // rs=1; en=0;
    *outk=3;  delay1(1);    *outk=2;  delay1(1);
    ptr++;
  } }
void init_lcd(){
  lcd_cmd(0x38);  // 8 bit bus & 2 rows function set
  lcd_cmd(0x0f);  // display on, cursor blink
  lcd_cmd(0x01);  // clear display
}
void lcd_cmd(volatile char cmd){
  *outf=cmd;  *outk=0;  delay1(1);  // rs=0; en=0;
  *outk=1;  delay1(1);   *outk=0;  delay1(1);
}
void init_port(){
  dirf=0x30; *dirf=0xff;  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}
void loop() {}
```

04 Display "ECEN Academy" on line2. Rotate right to left continuously :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  volatile char a;
  volatile long i;
  init_port();
  init_lcd();

while(1){
  lcd_cmd(0x0a);  // Display off cursor on
  lcd_cmd(0xd1);
  lcd_data("ECEN Academy");
  lcd_cmd(0x0c);  // Display on cursor off

  for(a=0; a<29; a++){
    lcd_cmd(0x18);
    for(i=0; i<100000; i++);
  }
  lcd_cmd(0x01);
 } }
void lcd_data(volatile char *ptr){
while(*ptr){
    *outf=*ptr;   *outk=2;  delay1(1);  // rs=1; en=0;
    *outk=3;  delay1(1);    *outk=2;  delay1(1);
    ptr++;
  } }
void init_lcd(){
  lcd_cmd(0x38);  // 8 bit bus & 2 rows function set
  lcd_cmd(0x0f);  // display on, cursor blink
  lcd_cmd(0x01);  // clear display
}
void lcd_cmd(volatile char cmd){
  *outf=cmd;  *outk=0;  delay1(1);  // rs=0; en=0;
  *outk=1;  delay1(1);   *outk=0;  delay1(1);
}
void init_port(){
  dirf=0x30; *dirf=0xff;  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}
void loop() {}
```

05 Display “Welcome to" on line1 in middle and “ECEN Academy” on line2. Rotate line2 right to left continuously :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  volatile short a;
  volatile long i;
  init_port();  init_lcd();

while(1){
  for(a=0xd0; a>=0xb5; a--){
    lcd_cmd(0x01);
    lcd_cmd(0x83);
    lcd_data("Welcome To");
    lcd_cmd(a);
    lcd_data("ECEN Academy");
    for(i=0; i<100000; i++);
  } } }

void lcd_data(volatile char *ptr){
while(*ptr){
    *outf=*ptr;    *outk=2;  delay1(1);
    *outk=3;  delay1(1);    *outk=2;  delay1(1);
    ptr++;
  }  }

void init_lcd(){
  lcd_cmd(0x38); lcd_cmd(0x0c); lcd_cmd(0x01);
}
void lcd_cmd(volatile char cmd){
  *outf=cmd;
  *outk=0;  delay1(1);
  *outk=1;  delay1(1);
  *outk=0;  delay1(1);
}
void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}
void loop() {}
```

06 Display “Welcome to" on line1. Rotate line1 right to left Continuously and “ECEN Academy” on line2 in Middle :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  volatile short a;
  volatile long i;
  init_port();  init_lcd();

while(1){
  for(a=0x9a; a>=0x80; a--){
    lcd_cmd(0x01);
    lcd_cmd(0x04);  // for left shift the cursor;
    lcd_cmd(a);
    lcd_data("oT emocleW");

    lcd_cmd(0xcd);
    lcd_data("ymedacA NECE");
    for(i=0; i<100000; i++);
  } } }

void lcd_data(volatile char *ptr){
while(*ptr){
    *outf=*ptr;    *outk=2;  delay1(1);
    *outk=3;  delay1(1);    *outk=2;  delay1(1);
    ptr++;
  }  }
void init_lcd(){
  lcd_cmd(0x38); lcd_cmd(0x0c); lcd_cmd(0x01);
}
void lcd_cmd(volatile char cmd){
  *outf=cmd;
  *outk=0;  delay1(1);
  *outk=1;  delay1(1);
  *outk=0;  delay1(1);
}
void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}
void loop() {}
```

<u>07 Display “Welcome to" on line1 in middle. “ECEN Academy” on line2 and rotate line2 left to right continuously :-</u>

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  volatile short a;
  volatile long i;
  init_port();  init_lcd();

while(1){
  for(a=0xc0; a<=0xdb; a++){
    lcd_cmd(0x01);
    lcd_cmd(0x83);
    lcd_data("Welcome To");

    lcd_cmd(0x04);  // for left shift the cursor;
    lcd_cmd(a);
    lcd_data("ymedacA NECE");
    for(i=0; i<100000; i++);
  } } }

void lcd_data(volatile char *ptr){
while(*ptr){
    *outf=*ptr;    *outk=2;  delay1(1);
    *outk=3;  delay1(1);    *outk=2;  delay1(1);
    ptr++;
  }  }

void init_lcd(){
  lcd_cmd(0x38); lcd_cmd(0x0c); lcd_cmd(0x01);
}
void lcd_cmd(volatile char cmd){
  *outf=cmd;
  *outk=0;  delay1(1);
  *outk=1;  delay1(1);
  *outk=0;  delay1(1);
}
void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}
void loop() {}
```

<u>08 Display “Welcome to" on line1. Rotate line1 left to right Continuously and “ECEN Academy” on line2 in Middle :-</u>

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  volatile short a;
  volatile long i;
  init_port();  init_lcd();

while(1){
  for(a=0x80; a<=0x9b; a++){
    lcd_cmd(0x01);
    lcd_cmd(0x04);  // for left shift the cursor;
    lcd_cmd(a);
    lcd_data("oT emocleW");

    lcd_cmd(0x06);  // for right shift the cursor
    lcd_cmd(0xc2);
    lcd_data("ECEN Academy");
    for(i=0; i<100000; i++);
  } } }

void lcd_data(volatile char *ptr){
while(*ptr){
    *outf=*ptr;    *outk=2;  delay1(1);
    *outk=3;  delay1(1);    *outk=2;  delay1(1);
    ptr++;
  }  }
void init_lcd(){
  lcd_cmd(0x38); lcd_cmd(0x0c); lcd_cmd(0x01);
}
void lcd_cmd(volatile char cmd){
  *outf=cmd;
  *outk=0;  delay1(1);
  *outk=1;  delay1(1);
  *outk=0;  delay1(1);
}
void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}
void loop() {}
```

09 Display “Welcome to" on line1. Rotate right to left continuously. “ECEN Academy" on line2. Rotate left to right continuously :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  volatile short a, b;
  volatile long i;
  init_port();  init_lcd();

while(1){
  for(a=0x9a, b=0xc0; a>=0x80; a--, b++){
    lcd_cmd(0x01);
    lcd_cmd(0x04);  // for left shift the cursor;
    lcd_cmd(a);
    lcd_data("oT emocleW");

    lcd_cmd(0x04);  // for left shift the cursor;
    lcd_cmd(b);
    lcd_data("ymedacA NECE");
    for(i=0; i<100000; i++);
  } } }

void lcd_data(volatile char *ptr){
while(*ptr){
    *outf=*ptr;    *outk=2;  delay1(1);
    *outk=3;  delay1(1);    *outk=2;  delay1(1);
    ptr++;
  }  }
void init_lcd(){
  lcd_cmd(0x38); lcd_cmd(0x0c); lcd_cmd(0x01);
}
void lcd_cmd(volatile char cmd){
  *outf=cmd;
  *outk=0;  delay1(1);
  *outk=1;  delay1(1);
  *outk=0;  delay1(1);
}
void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}
void loop() {}
```

10 Display “Welcome to" on line1. Rotate left to right continuously. “ECEN Academy" on line2. Rotate right to left continuously :-

```
volatile char *dirf, *outf, *dirk, *outk;
void init_port();
void delay1(int);
void init_lcd();
void lcd_cmd(volatile char cmd);
void lcd_data(volatile char *ptr);

void setup(){
  volatile short a, b;
  volatile long i;
  init_port();  init_lcd();

while(1){
  for(a=0x80, b=0xda; a<=0x9b; a++, b--){
    lcd_cmd(0x01);
    lcd_cmd(0x04);  // for left shift the cursor;
    lcd_cmd(a);
    lcd_data("oT emocleW");

    lcd_cmd(b);
    lcd_data("ymedacA NECE");
    for(i=0; i<100000; i++);
  } } }

void lcd_data(volatile char *ptr){
while(*ptr){
    *outf=*ptr;    *outk=2;  delay1(1);
    *outk=3;  delay1(1);    *outk=2;  delay1(1);
    ptr++;
  }  }
void init_lcd(){
  lcd_cmd(0x38); lcd_cmd(0x0c); lcd_cmd(0x01);
}
void lcd_cmd(volatile char cmd){
  *outf=cmd;
  *outk=0;  delay1(1);
  *outk=1;  delay1(1);
  *outk=0;  delay1(1);
}
void init_port(){
  dirf=0x30; *dirf=0xff;
  dirk=0x107; *dirk=0x03;
  outf=0x31; outk=0x108;
}
void delay1(volatile int time){
  volatile long i;
  while(time--)  for(i=0; i<100; i++);
}
void loop() {}
```